EXPERIMENT : 9

Performance Evaluation of OFDMA Resource-Block Allocation for Multi-User Communication.

OBJECTIVES :

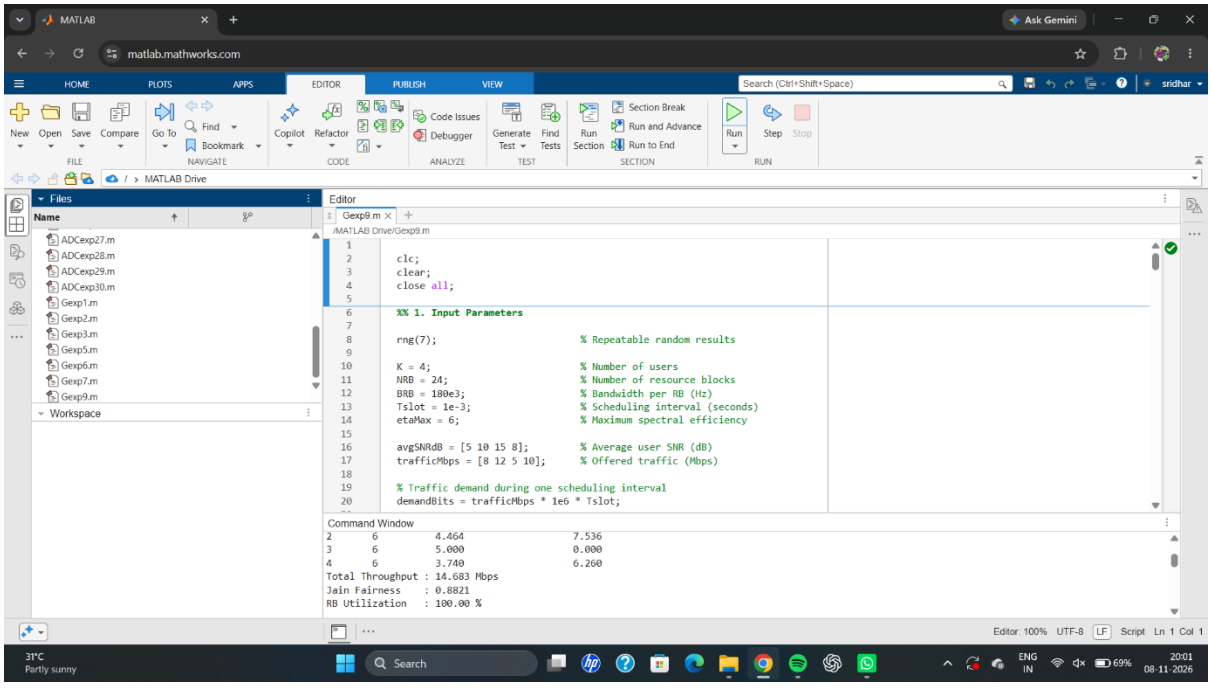
1. To represent a set of orthogonal OFDMA resource blocks and multiple users in MATLAB.

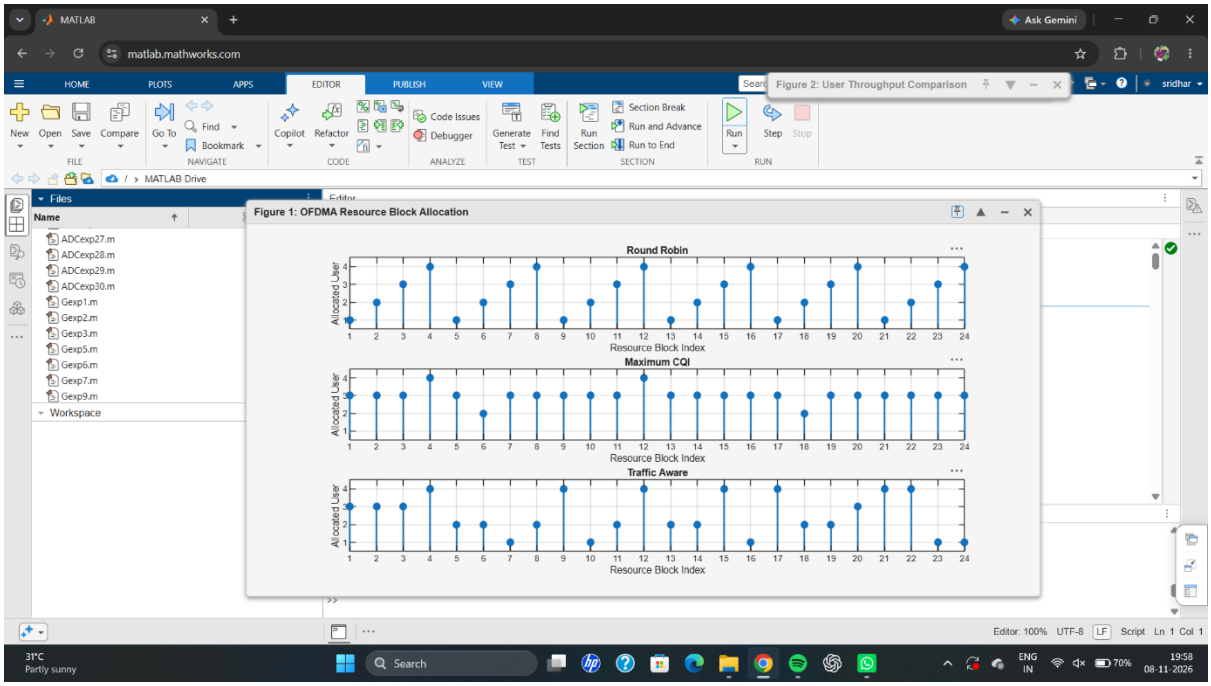
2. To allocate resource blocks using traffic-demand and channel-quality-based allocation rules.

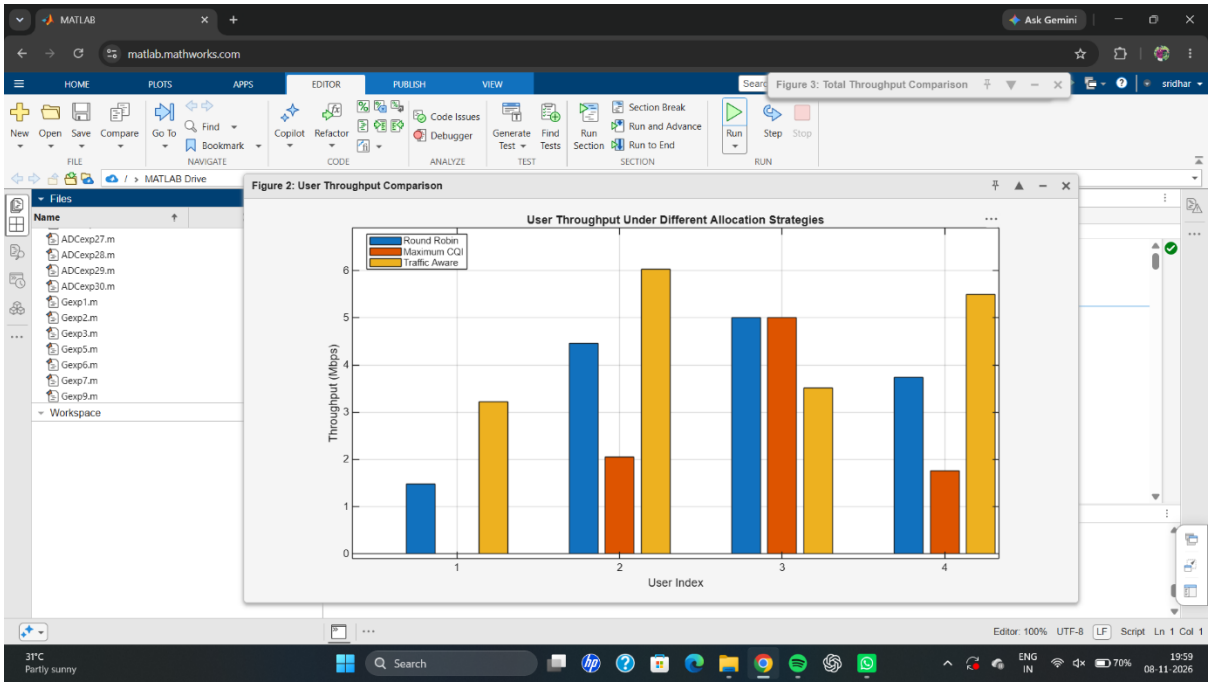
3. To compare user throughput, resource utilization and fairness under different allocation strategies.

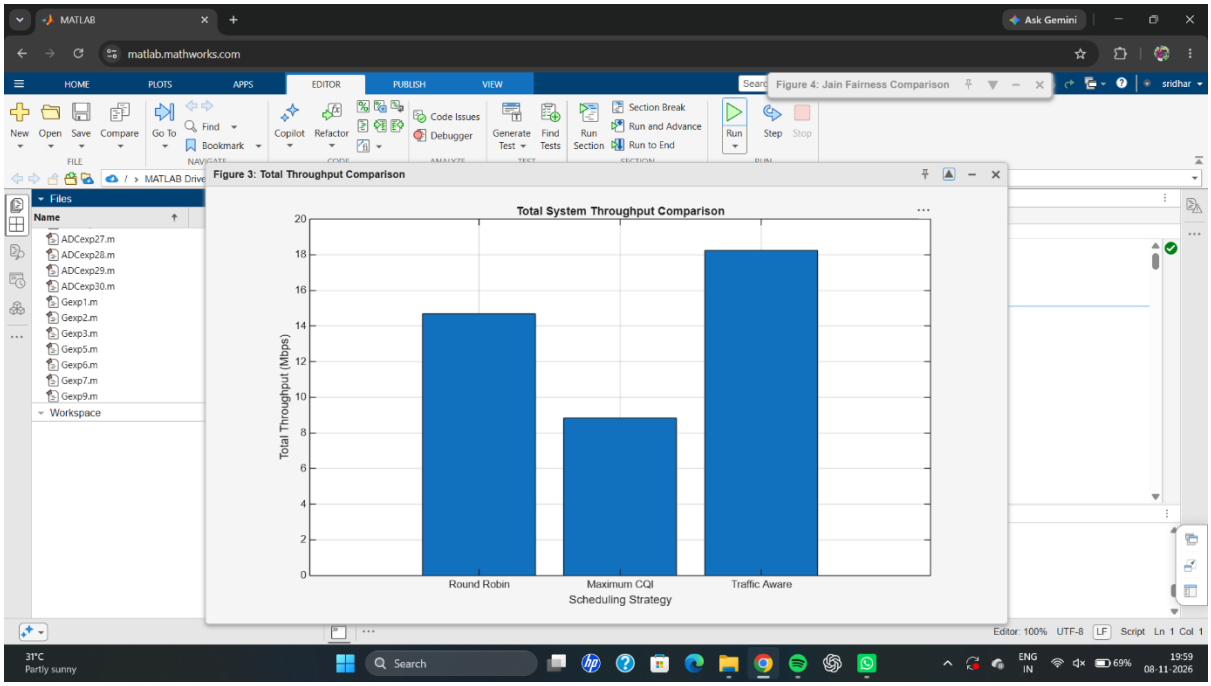
4. To explain how orthogonal resource allocation limits inter-user interference in multi-user communication.

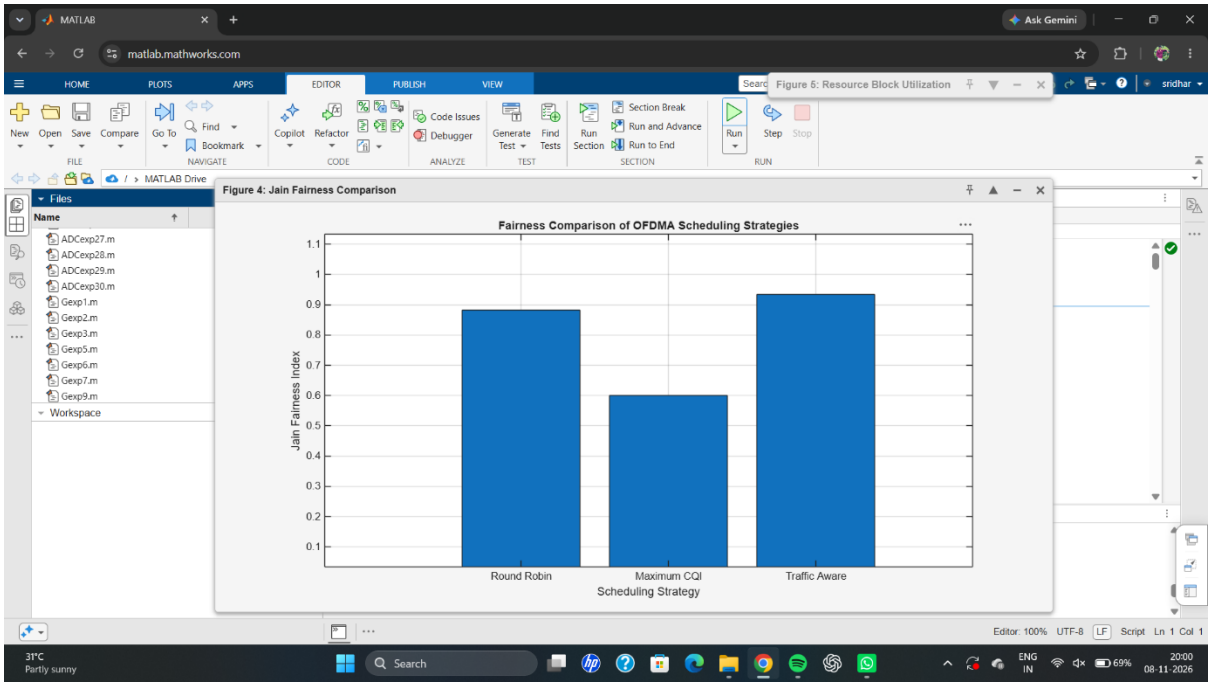
PROGRAME CODE :

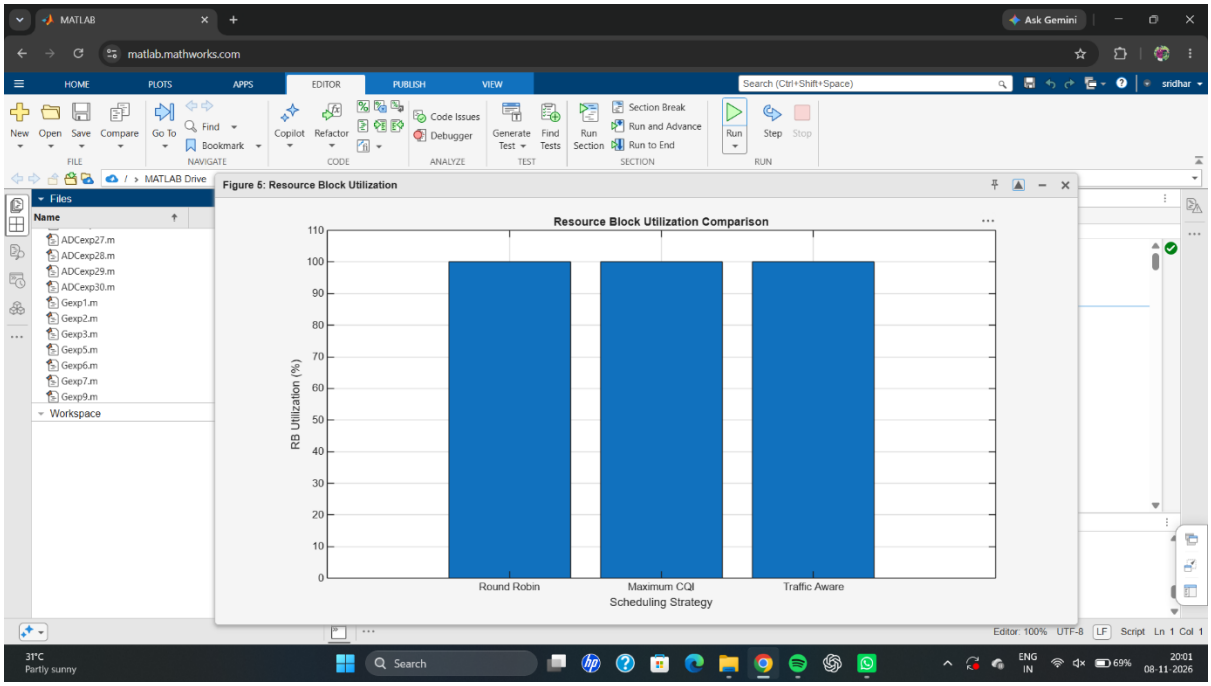



Figure 1: OFDMA Resource Block Allocation
Round Robin
Maximum CQI
Traffic Aware
Allocated User
Resource Block Index


Figure 2: User Throughput Comparison
User Throughput Under Different Allocation Strategies
Round Robin
Maximum CQI
Traffic Aware
Throughput (Mbps)
User Index


Figure 3: Total Throughput Comparison
Total System Throughput Comparison
Total Throughput (Mbps)
Round Robin
Maximum CQI
Traffic Aware
Scheduling Strategy


Figure 4: Jain Fairness Comparison
Fairness Comparison of OFDMA Scheduling Strategies
Jain Fairness Index
Round Robin
Maximum CQI
Traffic Aware
Scheduling Strategy



RESULT :

The MATLAB simulation successfully implements OFDMA resource block allocation using three scheduling strategies: Round Robin, Maximum CQI, and Traffic-Aware scheduling. Their performance is evaluated using user throughput, total system throughput, Jain's fairness index, and resource block utilization. The simulation demonstrates how different scheduling algorithms influence resource allocation, throughput, and fairness in wireless communication systems.